

Sec. 2.4: Writing direct proofs

Sec. 2.5: Proof by contrapositive

Direct proof of an implication: To prove an implication is true, we assume that the hypothesis is true, and use the hypothesis and other known true statements and definitions to deduce that the conclusion is true.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example 1. Consider the statement: “The sum of two odd integers is an even integer. ”

- (a) What is the hypothesis?
- (b) What is the conclusion?
- (c) Rewrite the statement in the format of “ $p \rightarrow q$ ”:

(d) *Proof.*

(1) Assume that .

We will show that .

(2) Since , .

Since , .

(3) Then, .

(4) Since , .

(5) Hence, .

□

Use the definitions! Recall definition in Sec. 2.1. Consider the following implications. What should be assume (and how to use it)? What should be proven at the end (the goal)?

- About parity of integers:

“If x is an even integer and y is an odd integer, then $2x + 5y$ is odd.”

Assume:

Goal:

- About rationality of numbers:

“If r is rational, then r^2 is also rational.”

Assume:

Goal:

- About divisibility of integers:

“If x, y are integers and $x \mid y$, then $x^2 \mid y^2$.”

Assume:

Goal:

Exercise 2. Consider the statement: “The sum of the square of an odd integer and the square of an even integer is an odd integer. ”

- What is the hypothesis?
- What is the conclusion?
- Rewrite the statement in the format of “ $p \rightarrow q$ ”:
- Prove the statement. (Hint: Follow the patterns in Example 1)

Prove by contrapositive: Since an implication is logically equivalent to its contrapositive, we may prove the contrapositive instead (if it’s easier!).

- The **contrapositive** of an implication $p \rightarrow q$ is .
- The contrapositive statement and the original statement are .

Example 3. Consider the statement:

“For any integers x and y , if $x - y$ is odd, then x is odd or y is odd. ”

- (a) What is the negation of the conclusion ($\neg q$)?
- (b) What is the negation of the hypothesis ($\neg p$)?
- (c) Rewrite the statement in the contrapositive format “ $\neg q \rightarrow \neg p$ ”:
- (d) Prove the contrapositive statement “ $\neg q \rightarrow \neg p$ ”.

When to use a direct proof vs. a proof by contrapositive

In principle, we choose the one with a hypothesis that is easier to work with.

- “For every integer n , if $n^2 - 2n + 7$ is even, then n is odd.”

	Hypothesis	Conclusion
Direct		
Contrapositive		

- “For every positive real number r , if r is irrational, then $\sqrt{r}$ is irrational. ”

	Hypothesis	Conclusion
Direct		
Contrapositive		

- Hypotheses that are usually easier to work with:
- Hypotheses that are usually harder to work with: